Allamaprabhu Ani

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SUMMARY

PhD student and lead developer of PhAST, a matrix-free differentiable PyTorch fracture solver. Experience in GPU finite elements, gradient-based inverse analysis, reproducible HPC workflows, and Excel-VBA aerospace analysis automation.

CORE COMPETENCIES

Scientific ML: Python, PyTorch, autograd, CUDA, NumPy/SciPy, neural operators, PINNs, inverse problems.
Mechanics: finite elements, fracture, Akantu, FEniCSx, COMSOL, Nastran/Patran, Abaqus, MATLAB.
Software: Slurm, Linux, Git, GitHub Actions, verification, benchmarking, technical documentation.

EXPERIENCE

PhD Student, Computational Mechanics and ML *Jul 2023 – present*
City St George's, University of London

- Develop and maintain PhAST's architecture, numerical kernels, documentation, and quasi-static and dynamic fracture workflows in PyTorch.
- Implemented scalar fracture-toughness recovery with autograd and checked gradients against finite differences; ongoing work explores spatial inverse problems and hybrid neural-operator workflows.
- Documented six fracture benchmark configurations and an approximately one-million-node NVIDIA A100 study; built reproducible Slurm workflows.
- Co-organised Mechanics and Machine Learning workshops in 2024 and 2025: venue booking, Qualtrics registration, speaker support, and hybrid broadcasting for an audience of around 70 researchers.

Visiting Researcher, Computational Solid Mechanics *2025*
LSMS Lab, EPFL (host: Prof. J.-F. Molinari)

- Lead author of the 2026 *Engineering Fracture Mechanics* review; coordinated contributions across City, EPFL, and the University of Florida.

Founder and Tool Developer *2020 – 2021*
Aeroknacks India Ltd

- Delivered Excel-VBA aerospace calculation tools using E.F. Bruhn, Michael Niu, Boeing Design Manuals, and ESDU: bending, lug strength, fastener loads, buckling, modal, and laminate analysis.

EDUCATION

PhD, Mechanical Engineering *Jul 2023 – present*
City St George's, University of London. Writing-up status.

MTech, Design and Manufacturing *2021 – 2023*
National Institute of Technology Silchar. CGPA 9.88/10; Best Student Award.

BE, Mechanical Engineering *2016 – 2020*
GM Institute of Technology, VTU. CGPA 8.47/10.

SELECTED PROJECTS

PhAST: lead developer of differentiable fracture simulation software. Source and documentation.
Mechanics and ML tutorials: hands-on UKACM Autumn School teaching support (2026). Course.

SELECTED PUBLICATIONS

Ani, A. et al. (2027). PhAST: A differentiable, matrix-free finite-element solver for dynamic phase-field fracture: Formulation, benchmarks, and inverse analysis. *Computer Methods in Applied Mechanics and Engineering* 464, 119547. DOI.

Ani, A.S. et al. (2026). Machine learning for computational fracture and damage mechanics: Status and perspectives. *Engineering Fracture Mechanics* 332, 111778. DOI.

AWARDS

Yeoman and Travelling Scholarship (2026); Dean's Award for Outstanding Teaching Support (2026); AFHEA (2024); Funded PhD Studentship (2023); Best Student Award (2023); Royal Aeronautical Society Member (2026).